

GrounUp Enterprise Specification

Phase 19 - Contract, Change Order, Claims & Commercial Management

Version 1.0.0 / Review-ready / Proposed lock: GES-P19-v1.0.0

1. Mission

Create a governed commercial-management platform that links contract obligations, notices, changes, pricing, schedule/cost impacts, claims, payments and evidence into a single auditable commercial record.

2. Architecture Principles

Contract-source authority; immutable approved versions; explicit approval authority; deadline awareness; segregation of duties; evidence lineage; cost/schedule/document integration; jurisdiction configurability; human-governed AI.

3. Contract & Clause Control

Contracts, amendments, exhibits and clauses are versioned. Obligations, notice requirements, payment/change mechanisms and responsible parties are linked back to the controlling source.

4. Change Management

Potential changes progress through scope, pricing, schedule impact, authorization, negotiation, approval/execution and reconciliation. T&M and force-account records retain daily backup and signatures.

5. Claims & Delay Records

Claims maintain chronology, cost/time components, positions and evidence. Delay, disruption, acceleration and differing-site-condition records link to Phase 18 schedules and Phase 15 evidence without automatically making legal conclusions.

6. Payment & Contract Value

Original value, approved/pending/forecast changes, payment applications, retainage, allowances, contingencies, backcharges and price adjustments reconcile into current commercial exposure.

7. Authority & Compliance

Binding notices, executed changes, waivers, settlements and commercial commitments require authorized human action and configured segregation-of-duties controls.

8. AI Commercial Intelligence

AI may extract clauses, flag potential notice requirements, classify changes, review backup completeness and surface commercial risk. AI cannot waive rights, send binding notices, settle claims, approve commitments or make legal conclusions.

9. Cross-Phase Integration

Phase 19 integrates controlled documents/RFI's/submittals, scheduling/delay evidence, estimating/cost/finance, procurement/subcontracts, field records, HSEQ references and enterprise analytics.

10. Commercial Digital Twin

The digital twin maintains current contract value, obligations, notices, changes, claims, payments, exposure, evidence, approvals and decision history.

11. Functional Domains

Commercial Management Core

Project commercial controls, contract hierarchy, commitments, notices, changes, claims, entitlements, approvals and audit.

Prime Contract Management

Owner/client agreements, amendments, exhibits, commercial terms, values, dates and obligations.

Subcontract Management

Subcontracts, scopes, exhibits, flow-downs, commitments, changes, notices and closeout.

Purchase Contract Commercials

Commercial terms linked to procurement commitments, allowances, escalation and delivery obligations.

Contract Clause Registry

Governed clauses, clause types, obligations, notice periods, payment terms, change mechanisms and references.

Obligation Management

Contractual obligations, owners, due dates, evidence, status, escalation and completion.

Notice Management

Notice events, required recipients, contractual deadlines, delivery evidence, reservations and follow-up.

Potential Change Management

Potential change events, source events, scope impacts, pricing status, schedule impacts and disposition.

Change Order Management

PCOs, owner changes, subcontract changes, change directives, approvals, revisions and execution.

Change Pricing

Labor, equipment, material, subcontract, trucking, burden, markup, tax, bond and allowance pricing.

T&M / Force Account

Daily tickets, labor/equipment/material usage, signatures, backup, rates and reconciliation.

Change Authorization

Proceed/no-proceed authority, field directives, not-to-exceed limits, approval matrices and audit.

Contract Value Control

Original value, approved changes, pending changes, forecast changes, revised value and exposure.

Claims Management

Claim events, theories/positions, chronology, cost/time components, evidence, review and status.

Delay & Disruption Records

Schedule-linked delay, disruption, acceleration, inefficiency and contemporaneous evidence.

Differing Site Conditions

Condition events, baseline information, observations, notice, investigation, cost/time impacts and evidence.

Scope & Design Change

Drawing/spec/RFI/submittal-driven scope changes, responsibility, quantity/cost/time impacts and traceability.

Payment & Billing Commercials

Applications for payment, pay items, retainage, stored materials, backcharges, deductions and certification.

Retainage & Holdbacks

Retainage rules, release milestones, withholding reasons, balances and approvals.

Backcharges & Deductions

Backcharge notices, support, responsibility, pricing, dispute status and settlement.

Allowances & Contingencies

Contract allowances, owner contingencies, drawdowns, transfers, remaining balances and approvals.

Escalation & Price Adjustment

Index/reference adjustments, fuel/material escalation, thresholds, calculations and contract basis.

Insurance, Bonds & Commercial Compliance

Commercial certificates, bonds, limits, expiration, required endorsements and evidence references.

Lien & Waiver Administration

Conditional/unconditional waivers, payment linkage, status, jurisdiction-configurable requirements and evidence.

Dispute Management

Issues, negotiations, mediation/arbitration/litigation references, settlement authority and privileged-data controls.

Commercial Forecasting

Pending/approved changes, claims exposure, contingency, margin-at-completion and cash/commercial forecasts.

Contract Closeout

Final changes, releases, waivers, claims disposition, final payment, retainage and commercial archive.

AI Commercial Intelligence

Evidence-grounded clause extraction, notice/deadline assistance, change classification, backup review and commercial risk recommendations.

Commercial Digital Twin

Current contract, obligation, change, claim, payment, exposure, evidence and decision state.

12. Controlled Artifact Summary

Artifact	Count
Requirements	420
Entities	195
Fields	320
Business Rules	85
Validation Rules	85
Workflows	51
Engines	53
Api Resources	68
Ui Surfaces	49
Reports	44
Kpis	54
Tests	420

13. KPI Registry

P19-KPI-001 - Notice On-Time %: Governed metric definition for Notice On-Time %.

P19-KPI-002 - Open Obligation Count: Governed metric definition for Open Obligation Count.

P19-KPI-003 - Overdue Obligation Count: Governed metric definition for Overdue Obligation Count.

P19-KPI-004 - Potential Change Aging: Governed metric definition for Potential Change Aging.

P19-KPI-005 - Change Approval Cycle Time: Governed metric definition for Change Approval Cycle Time.

P19-KPI-006 - Change Pricing Cycle Time: Governed metric definition for Change Pricing Cycle Time.

P19-KPI-007 - Change Conversion Rate: Governed metric definition for Change Conversion Rate.

P19-KPI-008 - Approved Change Value: Governed metric definition for Approved Change Value.

P19-KPI-009 - Pending Change Value: Governed metric definition for Pending Change Value.

P19-KPI-010 - Forecast Change Value: Governed metric definition for Forecast Change Value.

P19-KPI-011 - Contract Value Growth %: Governed metric definition for Contract Value Growth %.

P19-KPI-012 - T&M Ticket Approval %: Governed metric definition for T&M Ticket Approval %.

P19-KPI-013 - Unsigned T&M Ticket Count: Governed metric definition for Unsigned T&M Ticket Count.

P19-KPI-014 - NTE Exposure: Governed metric definition for NTE Exposure.

P19-KPI-015 - Change Gross Margin: Governed metric definition for Change Gross Margin.

P19-KPI-016 - Change Schedule Days: Governed metric definition for Change Schedule Days.

P19-KPI-017 - Claim Count: Governed metric definition for Claim Count.

P19-KPI-018 - Claim Value: Governed metric definition for Claim Value.

P19-KPI-019 - Claim Aging: Governed metric definition for Claim Aging.

P19-KPI-020 - Claim Recovery %: Governed metric definition for Claim Recovery %.

P19-KPI-021 - Claim Evidence Completeness: Governed metric definition for Claim Evidence Completeness.

P19-KPI-022 - Differing Site Condition Count: Governed metric definition for Differing Site Condition Count.

P19-KPI-023 - Delay Event Count: Governed metric definition for Delay Event Count.

P19-KPI-024 - Disruption Exposure: Governed metric definition for Disruption Exposure.

P19-KPI-025 - Acceleration Cost: Governed metric definition for Acceleration Cost.

P19-KPI-026 - Payment Application Cycle Time: Governed metric definition for Payment Application Cycle Time.

P19-KPI-027 - Retainage Outstanding: Governed metric definition for Retainage Outstanding.

P19-KPI-028 - Backcharge Outstanding: Governed metric definition for Backcharge Outstanding.

P19-KPI-029 - Allowance Remaining: Governed metric definition for Allowance Remaining.

P19-KPI-030 - Contingency Remaining: Governed metric definition for Contingency Remaining.

P19-KPI-031 - Price Adjustment Value: Governed metric definition for Price Adjustment Value.

P19-KPI-032 - Insurance Compliance %: Governed metric definition for Insurance Compliance %.

P19-KPI-033 - Bond Compliance %: Governed metric definition for Bond Compliance %.

P19-KPI-034 - Lien/Waiver Compliance %: Governed metric definition for Lien/Waiver Compliance %.

P19-KPI-035 - Commercial Exposure: Governed metric definition for Commercial Exposure.

P19-KPI-036 - Forecast Margin at Completion: Governed metric definition for Forecast Margin at Completion.

P19-KPI-037 - Commercial Forecast Variance: Governed metric definition for Commercial Forecast Variance.

P19-KPI-038 - Settlement Cycle Time: Governed metric definition for Settlement Cycle Time.

P19-KPI-039 - Closeout Commercial Aging: Governed metric definition for Closeout Commercial Aging.

P19-KPI-040 - Contract Clause Metadata Completeness: Governed metric definition for Contract Clause Metadata Completeness.

P19-KPI-041 - Notice Deadline Coverage: Governed metric definition for Notice Deadline Coverage.

P19-KPI-042 - Change Evidence Completeness: Governed metric definition for Change Evidence Completeness.

P19-KPI-043 - Claim Evidence Completeness: Governed metric definition for Claim Evidence Completeness.

P19-KPI-044 - Authority Exception Count: Governed metric definition for Authority Exception Count.

P19-KPI-045 - Segregation-of-Duties Exception Count: Governed metric definition for Segregation-of-Duties Exception Count.

P19-KPI-046 - AI Clause Extraction Acceptance: Governed metric definition for AI Clause Extraction Acceptance.

P19-KPI-047 - AI Notice Finding Acceptance: Governed metric definition for AI Notice Finding Acceptance.

P19-KPI-048 - AI Change Classification Acceptance: Governed metric definition for AI Change Classification Acceptance.

P19-KPI-049 - AI Backup Review Acceptance: Governed metric definition for AI Backup Review Acceptance.

P19-KPI-050 - AI Commercial Risk Acceptance: Governed metric definition for AI Commercial Risk Acceptance.

P19-KPI-051 - AI Evidence Coverage: Governed metric definition for AI Evidence Coverage.

P19-KPI-052 - Commercial Dashboard Adoption: Governed metric definition for Commercial Dashboard Adoption.

P19-KPI-053 - Commercial Digital Twin Freshness: Governed metric definition for Commercial Digital Twin Freshness.

P19-KPI-054 - Commercial Decision Cycle Time: Governed metric definition for Commercial Decision Cycle Time.

14. Engine Registry

P19-ENG-001 - Contract Version Engine (Deterministic): Provides contract version engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-002 - Clause Registry Engine (Deterministic): Provides clause registry engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-003 - Obligation Engine (Deterministic): Provides obligation engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-004 - Deadline/Notice Engine (Deterministic): Provides deadline/notice engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-005 - Notice Escalation Engine (Deterministic): Provides notice escalation engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-006 - Contract Authority Engine (Deterministic): Provides contract authority engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-007 - Approval Matrix Engine (Deterministic): Provides approval matrix engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-008 - Potential Change Engine (Deterministic): Provides potential change engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-009 - Change Classification Engine (Deterministic): Provides change classification engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-010 - Change Pricing Engine (Deterministic): Provides change pricing engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-011 - Labor Pricing Engine (Deterministic): Provides labor pricing engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-012 - Equipment Pricing Engine (Deterministic): Provides equipment pricing engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-013 - Material Pricing Engine (Deterministic): Provides material pricing engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-014 - Subcontract Pricing Engine (Deterministic): Provides subcontract pricing engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-015 - Trucking Pricing Engine (Deterministic): Provides trucking pricing engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-016 - Burden/Markup Engine (Deterministic): Provides burden/markup engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-017 - Tax/Bond Engine (Deterministic): Provides tax/bond engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-018 - T&M Reconciliation Engine (Deterministic): Provides t&m reconciliation engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-019 - Force Account Engine (Deterministic): Provides force account engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-020 - NTE Control Engine (Deterministic): Provides nte control engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-021 - Contract Value Engine (Deterministic): Provides contract value engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-022 - Pending Change Forecast Engine (Deterministic): Provides pending change forecast engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-023 - Commercial Exposure Engine (Deterministic): Provides commercial exposure engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-024 - Claim Chronology Engine (Deterministic): Provides claim chronology engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-025 - Claim Cost Engine (Deterministic): Provides claim cost engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-026 - Claim Time Engine (Deterministic): Provides claim time engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-027 - Delay Evidence Link Engine (Deterministic): Provides delay evidence link engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-028 - Disruption Evidence Engine (Deterministic): Provides disruption evidence engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-029 - Acceleration Cost Engine (Deterministic): Provides acceleration cost engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-030 - Differing Site Condition Engine (Deterministic): Provides differing site condition engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-031 - Scope Change Trace Engine (Deterministic): Provides scope change trace engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-032 - Payment Application Engine (Deterministic): Provides payment application engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-033 - Retainage Engine (Deterministic): Provides retainage engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-034 - Backcharge Engine (Deterministic): Provides backcharge engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-035 - Allowance Engine (Deterministic): Provides allowance engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-036 - Contingency Engine (Deterministic): Provides contingency engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-037 - Price Adjustment Engine (Deterministic): Provides price adjustment engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-038 - Escalation Index Engine (Deterministic): Provides escalation index engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-039 - Insurance/Bond Compliance Engine (Deterministic): Provides insurance/bond compliance engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-040 - Lien/Waiver Engine (Deterministic): Provides lien/waiver engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-041 - Commercial Forecast Engine (Deterministic): Provides commercial forecast engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-042 - Margin-at-Completion Engine (Deterministic): Provides margin-at-completion engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-043 - Settlement Authority Engine (Deterministic): Provides settlement authority engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-044 - Commercial Closeout Engine (Deterministic): Provides commercial closeout engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-045 - Commercial KPI Engine (Deterministic): Provides commercial kpi engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-046 - Clause Extraction Engine (AI-assisted): Provides clause extraction engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-047 - AI Notice Finding Engine (AI-assisted): Provides ai notice finding engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-048 - AI Change Classification Engine (AI-assisted): Provides ai change classification engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-049 - AI Backup Review Engine (AI-assisted): Provides ai backup review engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-050 - AI Commercial Risk Engine (AI-assisted): Provides ai commercial risk engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-051 - Evidence Retrieval Engine (AI-assisted): Provides evidence retrieval engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-052 - AI Confidence Engine (AI-assisted): Provides ai confidence engine using governed contract, cost, schedule, document and approval evidence.

P19-ENG-053 - Commercial Digital Twin Engine (AI-assisted): Provides commercial digital twin engine using governed contract, cost, schedule, document and approval evidence.

15. Workflow Registry

P19-WF-001 - Contract Intake & Setup: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-002 - Contract Version/Amendment: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-003 - Clause Extraction & Review: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-004 - Obligation Assignment: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-005 - Obligation Deadline Monitoring: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-006 - Notice Requirement Setup: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-007 - Notice Event Creation: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-008 - Notice Review & Delivery: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-009 - Potential Change Event: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-010 - Change Scope Review: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-011 - Change Pricing: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-012 - T&M Ticket Capture: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-013 - T&M Ticket Approval: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-014 - Change Schedule Impact Review: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-015 - Change Authorization: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-016 - Owner Change Order: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-017 - Subcontract Change Order: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-018 - Change Directive: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-019 - NTE Authorization: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-020 - Change Negotiation: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-021 - Change Approval & Execution: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-022 - Change Rejection/Close: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-023 - Contract Value Reconciliation: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-024 - Claim Event Intake: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-025 - Claim Chronology Build: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-026 - Claim Cost Build: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-027 - Claim Time Build: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-028 - Claim Evidence Review: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-029 - Claim Position Review: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-030 - Claim Submission: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-031 - Claim Negotiation: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-032 - Claim Settlement/Disposition: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-033 - Differing Site Condition: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-034 - Design/Scope Change Trace: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-035 - Delay/Disruption Evidence: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-036 - Acceleration Event: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-037 - Payment Application: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-038 - Retainage Management: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-039 - Backcharge: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-040 - Allowance/Contingency Draw: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-041 - Escalation/Price Adjustment: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-042 - Insurance/Bond Compliance: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-043 - Lien/Waiver Administration: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-044 - Commercial Forecast Update: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-045 - Dispute Escalation: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-046 - Settlement Authority: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-047 - Final Change Reconciliation: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-048 - Commercial Closeout: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-049 - AI Commercial Review: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-050 - Commercial Digital Twin Refresh: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

P19-WF-051 - Commercial Audit & Archive: Initiate > Validate > Price/Assess > Review > Authorize/Approve > Execute/Submit > Monitor > Close

16. Acceptance & Lock Gate

Each controlled requirement maps to a test case and implementation surface. Phase 19 remains review-ready until Commercial/Contracts, Project Controls, Finance, Operations, Legal/Contracts Review, Architecture, Security/Privacy, AI Governance, Data Governance, integration and UAT approvals are complete.